```
yl15687_nyu_edu@nyu-dataproc-m:~$ hdfs dfs -ls /user/$USER/hw1/q1_output
Found 2 items
-rw-r--r--   1 yl15687_nyu_edu yl15687_nyu_edu          0 2026-09-24 05:24 /user/yl15687_nyu_edu/hw1/q1_output/_SUCCESS
-rw-r--r--   1 yl15687_nyu_edu yl15687_nyu_edu        101 2026-09-24 05:24 /user/yl15687_nyu_edu/hw1/q1_output/part-00000
yl15687_nyu_edu@nyu-dataproc-m:~$ hdfs dfs -cat /user/$USER/hw1/q1_output/part-00000
Word    Frequency
the     17121
to      11622
of      10786
and     8814
a       7040
i       6480
in      5852
he      5054
that    4120
was     4004
```

```
yl15687_nyu_edu@nyu-dataproc-m:~$ hadoop jar /usr/lib/hadoop/hadoop-streaming-3.3.6.jar \
  -D mapreduce.job.reduces=1 \
  -files mapper1.py,reducer1.py \
  -mapper "python3 mapper1.py" \
  -reducer "python3 reducer1.py" \
  -input /user/$USER/hw1/input.txt \
  -output /user/$USER/hw1/q1_output
packageJobJar: [] [/usr/lib/hadoop/hadoop-streaming-3.3.6.jar] /tmp/streamjob17957563950007876837.jar tmpDir=null
2026-09-24 05:23:28,409 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at nyu-dataproc-m.c.hpc-dataproc-19b8.internal./192.168.1.70:8032
2026-09-24 05:23:28,592 INFO client.AHSProxy: Connecting to Application History server at nyu-dataproc-m.c.hpc-dataproc-19b8.internal./192.168.1.70:10200
2026-09-24 05:23:29,112 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at nyu-dataproc-m.c.hpc-dataproc-19b8.internal./192.168.1.70:8032
2026-09-24 05:23:29,113 INFO client.AHSProxy: Connecting to Application History server at nyu-dataproc-m.c.hpc-dataproc-19b8.internal./192.168.1.70:10200
2026-09-24 05:23:29,620 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/yl15687_nyu_edu/.staging/job_1788279331828_0990
2026-09-24 05:23:30,108 INFO mapred.FileInputFormat: Total input files to process : 1
2026-09-24 05:23:30,184 INFO mapreduce.JobSubmitter: number of splits:6
2026-09-24 05:23:30,374 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1788279331828_0990
2026-09-24 05:23:30,374 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-24 05:23:30,575 INFO conf.Configuration: resource-types.xml not found
2026-09-24 05:23:30,576 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-24 05:23:30,667 INFO impl.YarnClientImpl: Submitted application application_1788279331828_0990
2026-09-24 05:23:30,702 INFO mapreduce.Job: The url to track the job: http://nyu-dataproc-m.c.hpc-dataproc-19b8.internal.:8088/proxy/application_1788279331828_0990/
2026-09-24 05:23:30,703 INFO mapreduce.Job: Running job: job_1788279331828_0990
2026-09-24 05:23:40,824 INFO mapreduce.Job: Job job_1788279331828_0990 running in uber mode : false
2026-09-24 05:23:40,826 INFO mapreduce.Job:  map 0% reduce 0%
2026-09-24 05:23:53,943 INFO mapreduce.Job:  map 33% reduce 0%
2026-09-24 05:23:54,950 INFO mapreduce.Job:  map 100% reduce 0%
2026-09-24 05:24:01,990 INFO mapreduce.Job:  map 100% reduce 100%
2026-09-24 05:24:03,004 INFO mapreduce.Job: Job job_1788279331828_0990 completed successfully
2026-09-24 05:24:03,109 INFO mapreduce.Job: Counters: 55
        File System Counters
                FILE: Number of bytes read=3335258
                FILE: Number of bytes written=8724823
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=2037580
                HDFS: Number of bytes written=101
                HDFS: Number of read operations=23
```